

BIPLOB ROY

Data Engineer | Data Warehouse Specialist | AI Enthusiast
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PROFESSIONAL SUMMARY

Results-driven Data Engineer with expertise in data warehouse design, ETL pipelines, and large-scale migration. Proven experience building Snowflake OLAP systems and transforming complex transactional data into analytics-ready structures. Started in .NET backend development, now specializing in AI-driven solutions (ML.NET), delivering international healthcare and transport systems with end-to-end data architecture ownership.

CORE SKILLS

- Data Engineering: ETL Pipelines (SQL, Python, SSIS), Snowflake Schema Design, Fact/Dimension Modeling, OLTP → OLAP Transformation.
- Databases: MS SQL Server (OLAP, Performance Tuning), PostgreSQL, MongoDB.
- Backend & Distributed Systems: C#, ASP.NET MVC, Web API, Microservices, Kafka.
- AI & Tools: ML.NET, ONNX, Apache Spark, Git, Docker, SQL Server Agent, Airflow (basic).

PROFESSIONAL EXPERIENCE

Excel Technologies Ltd., Dhaka | Mar 2023 – Present
Data Engineer (Nov 2025 – Present)

- Designed SCRS (Smart Care Reporting Service) using Snowflake OLAP architecture on MS SQL Server, optimizing fact & dimension tables for high-performance reporting.
- Led end-to-end data migration pipeline (SmartCare Plus → Care Pro → SCRS) and developed custom ETL pipelines scheduled via SQL Server Agent.
- Designed PostgreSQL architecture for BTMS (Zambia) and built SQL functions & reporting APIs for real-time analytics.
- Developed an ML.NET conversational chatbot (R&D) for intelligent user interactions and automated ticket booking.

Junior Software Engineer (Mar 2023 – Oct 2025)

- Developed backend systems using ASP.NET MVC & Web API and optimized relational databases.

Digital Creation Plan, Khulna | Mar 2021 – Mar 2023
Junior Software Engineer

- Developed WPF applications and optimized database schemas for IoT-integrated real-time data systems.

Getco Online Ltd., Dhaka | May 2019 – Jan 2020
System Support Engineer

- Provided IT infrastructure support and ensured system reliability.

KEY PROJECTS

Tera Support – Multi-Microservice Platform & TSRS
Tech: .NET 8, Kafka, Docker, PostgreSQL, OLAP

- Built a high-performance Data Transport Pipeline syncing data every second from transactional databases into the TSRS OLAP Data Warehouse.
- Developed TSRS Reporting Service (.NET API) providing advanced dashboards, SLA filtering, and Excel exports.
- Designed Kafka event-driven architecture and an asynchronous Mail Service (SMTP/IMAP) for auto-ticket creation.

BTMS AI Chatbot (Conversational AI)
Tech: ML.NET, ONNX, .NET, SQL

- Developed a conversational AI chatbot using ML.NET and ONNX intent detection.
- Designed a dynamic retraining pipeline based on database intent datasets, and a 10-minute session inactivity manager.
- Implemented intent-based function calling to guide users through the full ticket purchase lifecycle.

SCRS – Smart Care Reporting Service (OLAP DW)
Tech: MS SQL Server, OLAP, Snowflake Schema, SQL Server Agent

- Designed centralized OLAP data warehouse and analytics-ready models for complex healthcare reporting.

Care Pro Data Migration Pipeline
Tech: MS SQL Server, SSIS, SQL

- Developed ETL logic (cleaning, transformation, validation) for a multi-system migration flow, unifying data for warehouse consumption.

RewardsPipeline (ETL System)
Tech: Python, MongoDB, MS SQL, Docker

- Built an automated ETL pipeline using pymongo and pyodbc; Dockerized for scalability and automated NoSQL to relational migration.

EDUCATION

B.Sc. in Computer Science & Engineering
Stamford University Bangladesh | CGPA: 3.26

CERTIFICATIONS

- IBM / Coursera Data Engineering Professional Certificate (Issued Apr 2026)
  (Includes Capstone, ML with Apache Spark, ETL & Data Pipelines, Data Warehouse Fundamentals, NoSQL, DBA)
- Generative AI: Elevate your Data Engineering Career – IBM (Apr 2026)
- BI Dashboards with IBM Cognos & Looker – IBM (Feb 2025)
- Data Science Specialization – Coursera (Completed)
- SQL (Intermediate & Basic) – HackerRank (Mar 2025)